

Name: _____

These questions assess your understanding of the material from Chapter(s) 27, 28 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

1. The primary mirror of an orbiting space telescope has a diameter of 2.0 m. One of the key advantages of operating in space is that it avoids atmospheric distortion, allowing the telescope to approach its diffraction-limited resolution. Assuming the telescope observes visible light with a wavelength of 600 nm, what is the minimum angular separation between two point sources that can be just resolved according to the Rayleigh criterion? Report your answer in arcseconds. Note: 1 degree is 60 arcminutes and 1 arcminute is 60 arcseconds.

$$D = 2.0 \text{ m}$$

$$\lambda = 600 \times 10^{-9} \text{ m}$$

$$\theta = 1.22 \frac{\lambda}{D}$$

$$= 1.22 \left(\frac{600 \times 10^{-9}}{2.0} \right) = 3.66 \times 10^{-7} \text{ rad} \left| \frac{180^\circ}{\pi \text{ rad}} \right| \frac{60 \text{ arcmin}}{1 \text{ degree}} \frac{60 \text{ arcseconds}}{1 \text{ arcmin}} = 0.075 \text{ arcseconds}$$

ANSWER: 0.075 arcseconds ✓

2. Suppose a space probe is moving away from Earth at a speed of 2.0×10^8 m/s. It sends a radio-wave message back to Earth at a frequency of 1.2 GHz. At what frequency is the message received on Earth?

$$v = 2.0 \times 10^8 \text{ m/s}$$

$$f_s = 1.2 \times 10^9 \text{ Hz}$$

$$f_o = ?$$

$$c = f \lambda$$

$$\lambda_s = \frac{c}{f}$$

$$= \frac{3.0 \times 10^8}{1.2 \times 10^9} = 0.25 \text{ m}$$

ANSWER: 5.36×10^8 Hz ✓

$$\lambda_o = \lambda_s \sqrt{\frac{1+u/c}{1-u/c}}$$

$$= (0.25) \sqrt{\frac{1+2/3}{1-2/3}}$$

$$= 0.56 \text{ m}$$

$$c = f \lambda$$

$$f = \frac{c}{\lambda}$$

$$= \frac{3.0 \times 10^8}{0.56} = 5.36 \times 10^8 \text{ Hz}$$

3. What angle would the axis of a polarizing filter need to make with the direction of polarized light of intensity 2.0 kW/m^2 to reduce the intensity to 0.75 kW/m^2 ?

$$I_o = 2.0 \frac{\text{kW}}{\text{m}^2} = 2.0 \times 10^3 \text{ W/m}^2$$

$$I = 0.75 \frac{\text{kW}}{\text{m}^2} = 0.75 \times 10^3 \text{ W/m}^2$$

$$I = I_o \cos^2 \theta$$

$$\frac{I}{I_o} = \cos^2 \theta$$

$$\theta = \cos^{-1} \left(\sqrt{\frac{I}{I_o}} \right)$$

$$= 5.22 \times 10^\circ$$

ANSWER: 52.2° ✓

+3

4. You shine a beam of white light through a diffraction grating with 5.5×10^3 lines per centimeter to a screen 1.2 m away. Find the angle for first-order diffraction of 550 nm light.

ANSWER: 17.6°

$$5.5 \times 10^3 \frac{\text{lines}}{\text{cm}} \left| \frac{100 \text{ cm}}{7 \text{ m}} \right| = 550,000$$

$$\frac{1}{N} = \frac{1}{550,000} = 1.82 \times 10^{-6} \text{ m} = d$$

$$\lambda = 550 \times 10^{-9} \text{ m}$$

$$m = 1$$

$$d \sin \theta = m \lambda$$

$$\theta = \sin^{-1} \left(\frac{m \lambda}{d} \right) = \sin^{-1} \left(\frac{(1)(550 \times 10^{-9})}{1.82 \times 10^{-6}} \right) = 17.6^\circ$$

5. Calculate the minimum thickness of an oil slick on water that appears blue when illuminated by white light perpendicular to its surface. Take the blue wavelength to be 450 nm, $n_{\text{oil}} = 1.46$ and $n_{\text{water}} = 1.33$.

ANSWER: $7.71 \times 10^{-8} \text{ m}$

$$n_{\text{air}} = 1 < n_{\text{oil}} = 1.46$$

Ray 1 shifted

$$n_{\text{oil}} = 1.46 > n_{\text{water}} = 1.33$$

Ray 2 not shifted

$$\lambda_n = \frac{\lambda}{n} = \frac{450 \times 10^{-9}}{1.46} = 3.08 \times 10^{-7} \text{ m}$$

$$\frac{1}{2} \times \lambda_n = 1.54 \times 10^{-7} \text{ m}$$

$$2t = 1.54 \times 10^{-7} \Rightarrow t = 7.71 \times 10^{-8} \text{ m}$$

6. Suppose a galaxy is moving away from Earth at a speed of $1.0 \times 10^8 \text{ m/s}$. It emits radio waves with a wavelength of 0.50 m. What wavelength would we detect on Earth?

ANSWER: 0.71 m

$$v = 1.0 \times 10^8 \text{ m/s}$$

$$\lambda_s = 0.50 \text{ m}$$

$$\lambda_{\text{obs}} = ?$$

$$\left| \begin{aligned} u/c &= \frac{1.0 \times 10^8}{3.0 \times 10^8} \\ &= \frac{1}{3} = 0.33 \end{aligned} \right|$$

$$\lambda_{\text{obs}} = \lambda_s \sqrt{\frac{1+u/c}{1-u/c}}$$

$$= 0.71 \text{ m}$$

+3

7. Visible light of wavelength 600 nm falls on a single slit and produces its second diffraction minimum at an angle of 24° relative to the incident direction of the light. What is the width of the slit?

$$\lambda = 600 \times 10^{-9} \text{ m}$$

$$m = 2$$

$$\theta = 24^\circ$$

$$D \sin \theta = m \lambda$$

$$D = \frac{m \lambda}{\sin \theta}$$

$$= \frac{(2)(600 \times 10^{-9})}{\sin(24^\circ)}$$

$$= 2.95 \times 10^{-6} \text{ m}$$

ANSWER: $2.95 \times 10^{-6} \text{ m}$ ✓

8. Suppose a cosmic ray colliding with a nucleus in Earth's upper atmosphere produces a muon that has a velocity equal to $1.1 \times 10^8 \text{ m/s}$. The muon then travels at constant velocity and lives for a duration of $1.2 \mu\text{s}$ in the muon's frame of reference. How long does the muon live as measured by an Earth-bound observer?

$$\gamma = \left(\sqrt{1 - \frac{v^2}{c^2}} \right)^{-1}$$

$$= \left(\sqrt{1 - \frac{(1.1 \times 10^8)^2}{(3.0 \times 10^8)^2}} \right)^{-1}$$

$$= 1.07$$

$$\Delta t = \gamma \Delta t_0$$

$$= (1.07)(1.2 \times 10^{-6})$$

$$= 1.29 \times 10^{-6} \text{ s}$$

ANSWER: $1.29 \times 10^{-6} \text{ s}$ ✓

9. The headlights of a car have a wavelength of about 550 nm and are 2.0-m apart. Assuming the eye's pupil diameter is 0.22 cm, what is the maximum distance at which the two headlights can be resolved? Use the diffraction limit (Rayleigh criterion) for your calculation.

$$\theta = 1.22 \frac{\lambda}{D}$$

$$= (1.22) \left(\frac{550 \times 10^{-9}}{0.22 \times 10^{-2}} \right)$$

$$= 3.05 \times 10^{-4} \text{ rad}$$

$$\Delta \theta = \frac{\Delta s}{r}$$

$$3.05 \times 10^{-4} = \frac{2.0}{r}$$

$$r = 6.56 \times 10^3 \text{ m}$$

ANSWER: $6.56 \times 10^3 \text{ m}$ ✓

+3

10. Suppose you pass light from a laser through two slits separated by 0.014 mm and find that the third bright line on a screen is formed at an angle of 30° relative to the incident beam. What is the wavelength of the light?

ANSWER: $2.33 \times 10^{-6} \text{ m}$

$$d = 0.014 \times 10^{-3} \text{ m}$$

$$m = 3$$

$$\theta = 30^\circ$$

$$d \sin \theta = m \lambda$$

$$\lambda = \frac{d \sin \theta}{m}$$

$$= \frac{(0.014 \times 10^{-3})(\sin(30^\circ))}{3} = 2.33 \times 10^{-6} \text{ m}$$

11. Calculate the energy that is released upon the complete annihilation of a 6.3 g object.

ANSWER: $5.67 \times 10^{14} \text{ J}$

$$\begin{aligned} E &= mc^2 \\ &= (6.3 \times 10^{-3})(3.0 \times 10^8)^2 \\ &= 5.67 \times 10^{14} \text{ J} \end{aligned}$$

12. Suppose an astronaut is traveling at $1.7 \times 10^8 \text{ m/s}$ from Earth to Alpha Centauri, which is a star system 2.5 ly away as measured by an Earth-bound observer. How far apart are Earth and Alpha Centauri as measured by the astronaut? Note: ly is lightyear, the distance light travels in one year.

ANSWER: 2.06 ly

$$L = L_0 \gamma^{-1}$$

$$\gamma = \left(\sqrt{1 - \frac{v^2}{c^2}} \right)^{-1}$$

$$= \left(\sqrt{1 - \frac{(1.7 \times 10^8)^2}{(3.0 \times 10^8)^2}} \right)^{-1}$$

$$= 1.21$$

$$L = (2.5)(1.21)^{-1}$$

$$= 2.06 \text{ ly}$$

+3

PHY202 Exam 7 on chapter(s): 27, 28

1. $\theta = 0.0755 \text{ arcseconds} \therefore \theta = 1.22 \frac{\lambda}{D}$
(Chapter 27, Section (6) Limits of Resolution - cple_ch27_ex5)
2. $f = 5.37 \times 10^8 \text{ Hz} \therefore \lambda_{obs} = \lambda_s \sqrt{\frac{1+\frac{u}{c}}{1-\frac{u}{c}}}, c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = \frac{E}{B} = f\lambda$
(Chapter 28, Section (4) Relativistic Addition of Velocities - cple_ch28_ex5b)
3. $\theta = 52.2^\circ \therefore I = I_0 \cos^2 \theta$
(Chapter 27, Section (8) Polarization - cple_ch27_p87)
4. $\theta = 17.6^\circ \therefore d \sin \theta = m\lambda$
(Chapter 27, Section (4) Multiple-Slit Diffraction - cple_ch27_ex3)
5. $t = 7.71 \times 10^{-8} \text{ m} \therefore 2t = \left(m + \frac{1}{2}\right) \lambda_n, n_{\text{air}} < n_{\text{oil}} > n_{\text{water}}$
(Chapter 27, Section (7) Thin-Film Interference - cple_ch27_p72)
6. $\lambda_{obs} = 0.707 \text{ m} \therefore \lambda_{obs} = \lambda_s \sqrt{\frac{1+\frac{u}{c}}{1-\frac{u}{c}}}$
(Chapter 28, Section (4) Relativistic Addition of Velocities - cple_ch28_ex5)
7. $D = 2.95 \times 10^{-6} \text{ m} \therefore D \sin \theta = m\lambda$
(Chapter 27, Section (5) Single-Slit Diffraction - cple_ch27_ex4)
8. $\Delta t = 1.29 \times 10^{-6} \text{ s} \therefore \gamma = \left(\sqrt{1 - \frac{v^2}{c^2}}\right)^{-1}, \Delta t = \gamma \Delta t_0$
(Chapter 28, Section (2) Simultaneity and Time Dilation - cple_ch28_ex1)
9. $x = 6.56 \times 10^3 \text{ m} \therefore \theta = 1.22 \frac{\lambda}{D}, \Delta \theta = \frac{\Delta s}{r}, \Delta s \approx 2.0 \text{ m for small angles}$
(Chapter 27, Section (6) Limits of Resolution - cple_ch27_p66)
10. $\lambda = 2.33 \times 10^{-6} \text{ m} \therefore d \sin \theta = m\lambda$
(Chapter 27, Section (3) Young's Double-Slit Experiment - cple_ch27_ex1)
11. $E = 5.67 \times 10^{14} \text{ J} \therefore E_0 = mc^2$
(Chapter 28, Section (6) Relativistic Energy - cple_ch28_ex6)
12. $L = 2.06 \text{ ly} \therefore L = L_0 \gamma^{-1}$
(Chapter 28, Section (3) Length Contraction - cple_ch28_ex2)